Model Diversity and Shared Blind Spots

An Efficient-Market Perspective on Combining Models

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- A provider can sell a useful combination as a new service. Competition may therefore absorb inexpensive improvements into the services already available.
- When accurate models cannot be improved by simple voting, their errors must concentrate on a common set of questions. More models need not supply complementary judgments.

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This also matters when we assemble several models into an oversight panel.

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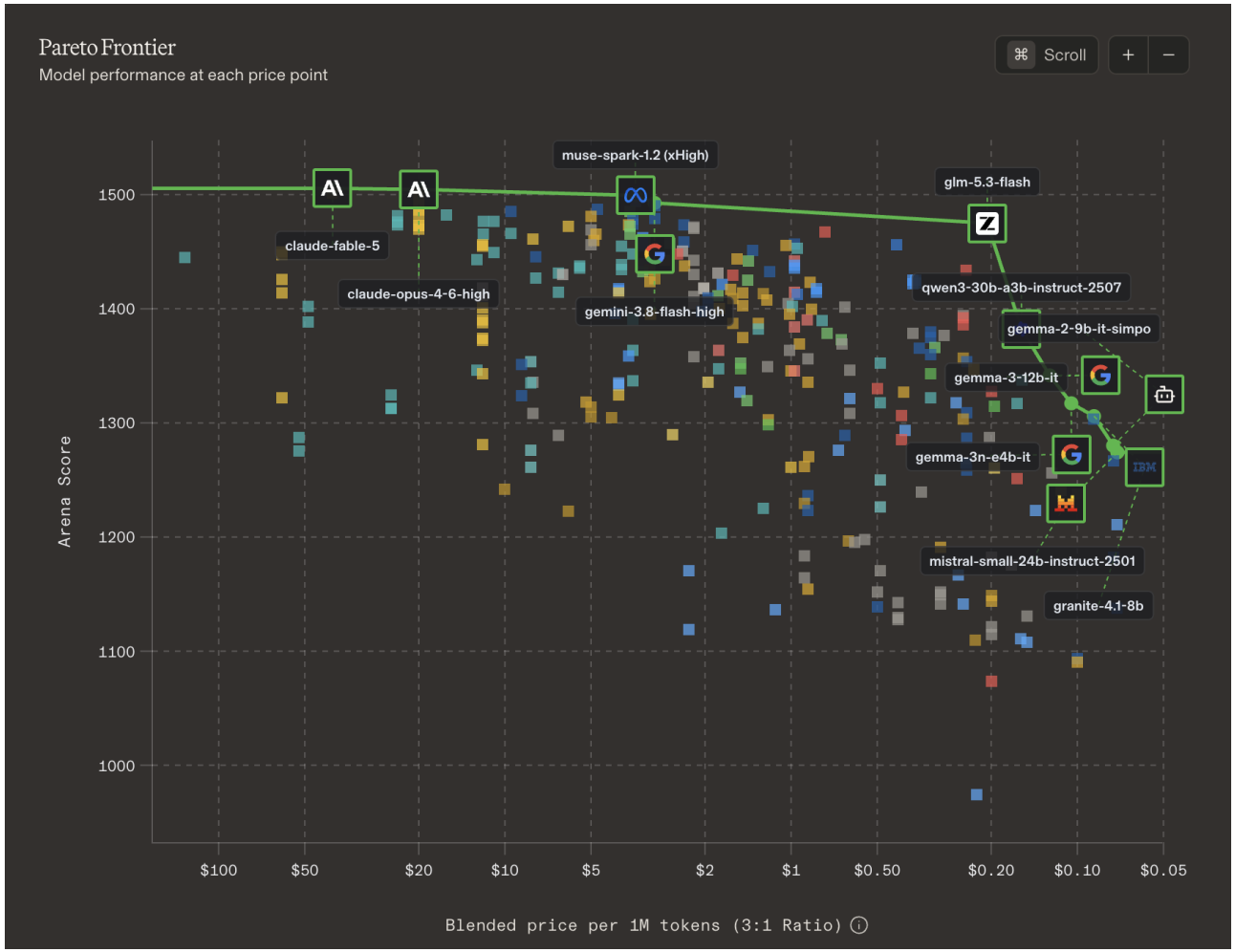
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The individual accuracies are identical in the two examples. What matters for voting is how the errors occur together.

What Performance Can We Buy at a Given Price?



Higher is better; cheaper is to the right. Source: Arena.

The *Pareto frontier* consists of models for which no other listed model is at least as good and no more expensive, with one strict improvement.

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Efficient-market hypothesis for AI

In a mature market, competition should tend to make simple, inexpensive ensemble improvements available as services buyers can already purchase.

This is a motivation for our hypothesis, not a theorem about market dynamics. The pressure is strongest for economically valuable tasks.

Model: Binary Answers on a Fixed Population

For the analysis, replace Arena score by accuracy on a binary prediction task. Let $X \sim \mu$ be a question from a finite set $\mathcal{X}$, omitting questions with probability zero. Each input has a fixed, correct binary label

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The models' answers are fixed; the random object is the question.
For oversight, an input is a proposal: its correct label is $+1$ if it does not harm the principal and -1 if it does.

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$$\alpha(B) := \min_{\substack{h \in \mathcal{M} \\ c(h) \le B}} \operatorname{err}(h).$$

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The better constant makes error at most $1/2$, so

$$0 < \alpha \le \frac{1}{2}.$$

Threshold Votes Count Positive Answers

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For each threshold $q \in \{1, \ldots, k\}$, define the vote

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The threshold stays fixed across questions. We idealize counting votes as free, so every v_q costs at most B.

A Panel Can Require a Fixed Number of Approvals

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Error counts both harmful approvals and non-harmful rejections. This is the accuracy question we study today.

Assume No Threshold Vote Beats the Budget Benchmark

Recall that $\alpha = \alpha(B)$ is the smallest error of an available service at budget B.

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Theorem: How Often Can Any Model Be Wrong?

Write

$$E_j := \{x : h_j(x) \neq y(x)\}, \qquad U := \bigcup_{j=1}^{k} E_j.$$

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Theorem (Error concentration)

If every threshold vote of the jointly affordable models has error at least $\alpha > 0$, then

$$\boxed{\mathbb{P}(U) \leq 2\alpha + \sum_{j=1}^{k} \delta_j.}$$

Threshold Votes Preserve the Error Count

Fix a question x and let $r = A(x)$ be its number of positive answers. The thresholds $q = 1, \ldots, r$ return $+1$; the other $k - r$ thresholds return -1.

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Whichever label is correct, both collections contain the same number of wrong answers on this question. Averaging this equality over questions gives

$$\boxed{\sum_{q=1}^{k} \mathrm{err}(v_q) = \sum_{j=1}^{k} \mathrm{err}(h_j).}$$

The Two Extreme Thresholds Detect Every Error

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If $y(x) = -1$ and some model is wrong, at least one model answers $+1$. Then $v_1(x) = +1$ is wrong.

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If $y(x) = -1$ and some model is wrong, at least one model answers $+1$. Then $v_1(x) = +1$ is wrong.
Thus whenever $x \in U$, at least one of these two votes is wrong:

$$U \subseteq \{x : v_1(x) \neq y(x)\} \cup \{x : v_k(x) \neq y(x)\}.$$

The union bound gives

$$\boxed{\mathbb{P}(U) \leq \mathrm{err}(v_1) + \mathrm{err}(v_k).}$$

The Interior Thresholds Complete the Bound

The bound from the two extreme thresholds and the equality of total errors give

$$\begin{aligned}\mathbb{P}(U) &\leq \mathrm{err}(v_1) + \mathrm{err}(v_k) \\ &= \sum_{j=1}^{k} \mathrm{err}(h_j) - \sum_{q=2}^{k-1} \mathrm{err}(v_q).\end{aligned}$$

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Each of the $k-2$ interior thresholds has error at least α. Using $\mathrm{err}(h_j) = \alpha + \delta_j$, we obtain

$$\begin{aligned}\mathbb{P}(U) &\leq \sum_{j} \mathrm{err}(h_j) - (k-2)\alpha \\ &= k\alpha + \sum_{j} \delta_j - (k-2)\alpha.\end{aligned}$$

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For five models and $\alpha = 0.1$, all errors lie within at most 20% of questions. Every model is correct on the remaining 80% or more.

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Equivalently, if $\mathbb{P}(U) > 2\alpha + \sum_j \delta_j$, at least one threshold vote must beat the best service at budget B. Sufficiently dispersed errors would allow an improvement.

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Both terms are nonnegative. Small excess error requires both good performance for the model's price and little improvement from spending the full budget.
When the frontier improves sharply with spending, the error-concentration bound can be weak even if every constituent is best at its own price.

Failures Occur Together on Average

Continue with $\text{err}(h_j) = \alpha$ for every model. The questions with any failure form $U = \bigcup_j E_j$, with $\mathbb{P}(U) \leq 2\alpha$. For each model,

$$\mathbb{P}(E_j \mid U) = \frac{\mathbb{P}(E_j)}{\mathbb{P}(U)} = \frac{\alpha}{\mathbb{P}(U)} \geq \frac{1}{2}.$$

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When any model fails, at least half fail *on average*. The models that fail can differ across questions.

Five Models Can Attain the Factor of Two

Consider this population of questions. Each row gives its probability, the correct label, and the models' answers.

Probability	Y	h_1	h_2	h_3	h_4	h_5
0.4	$+1$	$+$	$+$	$+$	$+$	$+$
0.4	-1	$-$	$-$	$-$	$-$	$-$
0.1	$+1$	$+$	$+$	$-$	$-$	$-$
0.1	-1	$+$	$+$	$-$	$-$	$-$

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Models h_1, h_2 err only on the last row. Models h_3, h_4, h_5 err only on the third row. Thus every model has error 0.1, but

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Could a voting rule improve on 0.1? The last two rows give it exactly the same answers.

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Any fixed rule using only these answers makes the same prediction on both rows. It is wrong on at least one row, giving error at least 0.1.

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Any fixed rule using only these answers makes the same prediction on both rows. It is wrong on at least one row, giving error at least 0.1.

Take a market containing these five jointly affordable models and the constants. Then $\alpha = 0.1$, no threshold vote improves on it, and $\mathbb{P}(U) = 2\alpha$. The bound is tight.

The Same Answers Conceal Opposite Correct Labels

On the last 20% of questions, the available answers are

Probability	Y	h_1	h_2	h_3	h_4	h_5
0.1	$+1$	$+$	$+$	$-$	$-$	$-$
0.1	-1	$+$	$+$	$-$	$-$	$-$

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The models' mistakes concentrate in a small common region, even though h_1 and h_3 never fail on the same question.

Summary

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Summary

- Competition gives a reason that simple ensemble improvements might already be available as services. Compare a new combination with what buyers can get at its *total* cost.
- If accurate models cannot improve on that benchmark by threshold voting, their errors concentrate on a small common set of questions. The bound depends on their excess errors at the full budget.
- Buying more models does not automatically buy complementary judgments. An oversight mechanism can use such complementarity, but establishing it is a separate design problem.

Optional: Averaging Probability Forecasts

The same market question applies when models report probabilities. Now a forecast is a vector

$$p(x) = (p_-(x), p_+(x)).$$

Its coordinates are nonnegative and sum to one. They give the reported probabilities of labels $-1, +1$.

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Two forecasts can be combined by reporting

$$\bar{p}(x) := \frac{p_1(x) + p_2(x)}{2}.$$

How different can p_1, p_2 be if their average cannot improve on the best available service?

No Improvement by Averaging Bounds Disagreement

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Averaging is free. Define their excess losses

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Theorem (Forecast agreement)

Under these assumptions,

$$\boxed{\mathbb{E}\|p_1(X) - p_2(X)\|_2^2 \leq 2(\varepsilon_1 + \varepsilon_2).}$$

The proof will identify exactly how much averaging reduces Brier loss.

The Exact Gain from Averaging Forecasts

Fix a question and write $a = p_1(x) - e_{y(x)}$, $b = p_2(x) - e_{y(x)}$ for the two forecast errors. Their average is the error of $\bar{p}$.

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$$\frac{\|a\|_2^2 + \|b\|_2^2}{2} - \left\| \frac{a+b}{2} \right\|_2^2 = \frac{\|a\|_2^2 + \|b\|_2^2 - 2\langle a, b \rangle}{4}$$
$$= \frac{1}{4}\|a - b\|_2^2.$$

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The squared disagreement measures the improvement over the *average loss of the two models*.

Forecasts Near the Benchmark Must Be Close

Using $L(p_j) = \beta + \varepsilon_j$ in the averaging identity gives

$$L(\bar{p}) = \beta + \frac{\varepsilon_1 + \varepsilon_2}{2} - \frac{1}{4}\mathbb{E}\|p_1(X) - p_2(X)\|_2^2.$$

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The no-improvement hypothesis $L(\bar{p}) \geq \beta$ therefore implies

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At zero excess loss, the forecasts agree on every positive-probability question, even if the benchmark loss β is large.

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At zero excess loss, the forecasts agree on every positive-probability question, even if the benchmark loss β is large. Substantial disagreement would let us beat the benchmark by averaging these near-benchmark forecasts.

Thanks!

See you next class!

References

- Surbhi Goel, Meena Jagadeesan, Michael Kearns, Sophia Pi, and Aaron Roth. *Ensemble Arbitrage and the Efficient Market Hypothesis for AI*. Unpublished manuscript, 2026. Section 3.2 gives the threshold-voting proof and tight example; Appendix A gives the market application for Brier loss.
- Arena. Text leaderboard, Pareto view. Cost–performance screenshot supplied for this lecture.

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